Summary and Recommendations

1. Objective

The objective of this analysis is to understand customer churn behaviour in a telecom dataset by examining how different services (internet type, security, support, streaming, etc.) influence customer retention. The goal is to identify key factors driving churn and provide actionable insights to reduce customer attrition.

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2. Key Insights

- Overall churn rate is approximately **26–27%**, indicating that about **1 in 4 customers leave the service**.
- Customers using **Fiber Optic Internet** have the highest churn rate (**~45–50%**), compared to DSL (**~20–25%**) and no internet (**<10%**).
- Lack of add-on services such as **Online Security, Tech Support, and Device Protection** leads to significantly higher churn (**~40–50%**).
- Customers with these add-on services show much lower churn (**~15–25%**), highlighting their importance in retention.
- Basic services like **PhoneService and MultipleLines** have minimal impact on churn behavior.
- Streaming services (**TV/Movies**) show moderate influence, with slightly lower churn among active users (**~25–30%**) compared to non-users (**~30–35%**).

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3. Executive Summary

This analysis highlights key drivers of customer churn in the telecom sector using visual exploration of service usage patterns. The findings reveal that churn is significantly influenced by the type of internet service and the availability of value-added services.

Customers subscribed to fiber optic internet exhibit the highest churn rates, making them the most critical segment for retention strategies. Additionally, the absence of services such as online security, tech support, and device protection strongly correlates with higher churn, suggesting that customers value bundled and supportive service offerings.

On the other hand, basic telecom services like phone connectivity and multiple lines do not significantly impact churn, while streaming services contribute moderately to customer engagement and retention.

Overall, improving service quality for high-risk segments and promoting add-on services can substantially reduce churn and enhance customer lifetime value.

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4. Visualizations and Insights

The analysis uses multiple **count plots (categorical vs churn)** to compare customer distribution:

- **PhoneService & MultipleLines:**
  Show high customer counts but relatively stable churn distribution, indicating low influence on churn.
- **InternetService:**
  Fiber optic users show visibly higher churn counts compared to DSL and no internet users, making it a major churn driver.
- **OnlineSecurity, TechSupport, DeviceProtection:**
  Clear visual gap where customers without these services have significantly higher churn bars compared to those who have them.
- **OnlineBackup:**
  Customers without backup services show higher churn, though the impact is slightly less than security/support services.
- **StreamingTV & StreamingMovies:**
  Moderate differences in churn, with slightly better retention among users of these services.

These visualizations clearly highlight that **service engagement and value-added features directly influence customer retention**.

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5. Recommendations

- **Target Fiber Optic Customers:**
  Investigate pricing, service quality, and customer experience to reduce high churn in this segment.
- **Promote Add-On Services:**
  Bundle **Online Security, Tech Support, and Device Protection** with core plans to improve retention.
- **Upselling Strategy:**
  Encourage customers to adopt additional services (backup, streaming) to increase engagement.
- **Customer Segmentation:**
  Identify high-risk customers (no add-ons + fiber users) and apply targeted retention campaigns.
- **Improve Customer Experience:**
  Focus on proactive support, better service reliability, and personalized offers to reduce churn.